CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M. Tech)
Semester – II University Examination May 2011
M. Tech. (CAD/CAM) (Theory)
ME – 707 Computer Aided Manufacturing

Date: 16.05.2011, Monday   Time: 10:00 a.m. to 01:00 p.m.   Maximum Marks: 70

Instructions:

1. Figures to the right indicate *full* marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION- I

Q1	**(a)**	Explain the factors considered while selecting components for machining on CNC machine tools.	**04**
	(b)	Explain Computer Assisted Programming – APT stating illustration.	**03**
	(c)	With neat schematic diagram, explain laminated object manufacturing process of rapid prototyping.	**04**
Q2	**(a)**	Explain with drawing Maximum Material Condition Principal (MMC).	**04**
	(b)	Explain types of references /datum's used for programming in CNC milling & lathe.	**04**
	(c)	Explain with sketch 3 axis and 5 axis machining stating illustrations.	**04**
		OR	
Q2	**(a)**	Explain fundamental rules for GD & T.	**04**
	(b)	Draw a sketch of horizontal spindle machining center with a rotary table and specify all the axes on it. State the rules of axes designation followed.	**04**
	(c)	List and explain the types of control based on positioning and motion of CNC system.	**04**
Q3	**(a)**	Write manual part program using absolute mode for profile turning a component as shown in figure 1. Sketch tool path showing work piece zero clearly. Assume speed, feed and necessary data stating clearly.	**06**
	(b)	Explain the circular interpolation writing statements, showing motions and citing illustrations.	**03**
	(c)	Write Threading Canned cycles and draw figure showing tool motions.	**03**
		OR	
Q3	**(a)**	Write a manual part program for edge milling a low-carbon steel rectangular component of size 120 x 80 x 10mm. Select and mention workpiece zero, start point and sketch tool path.	**06**
	(b)	With the help of flowchart, explain the procedure to be followed for generating NC part program using CAM software.	**03**
	(c)	Write Drilling Canned cycles and draw figure showing tool motions.	**03**

P.T.O

I

SECTION - II

Q4 (a) Discuss the IBM concept of CIM. **05**

(b) Form the part families for the following machine matrix using Direct Cluster algorithm. O indicates the required operation. **06**

	Parts								
Operations	1	2	3	4	5	6	7	8	9
A	O		O			O		O	
B		O			O		O		
C	O		O			O			
D	O					O		O	
E		O			O		O		
F				O					O
G				O					O

Q5 (a) Explain the process of Image processing and image analysis in machine vision. **04**

(b) With the help of a typical flexible manufacturing cell explain the working of FMS. **05**

(c) Enlist the design and manufacturing attributes used in GT coding. **03**

OR

Q5 (a) Explain Group Technology stating objectives and the two hurdles in implementing it **03**

(b) What is co-ordinate metrology? Why it is applied? Explain any one inspection instrument using co-ordinate metrology. **05**

(c) What are different types of FMS layouts? Explain. **04**

Q6 (a) Discuss the automated features of Turning centre **05**

(b) Explain the Opitz classification system showing basic structure and stating applications. **04**

(c) Explain how electronic inspection instruments are superior then conventional instruments? **03**

OR

Q6 (a) Define flexibility. What are types of flexibilities? Explain any two in detail. **05**

(b) With neat sketch explain the major elements of CIM. **03**

(c) Explain the principle and working of scanning laser system. **04**

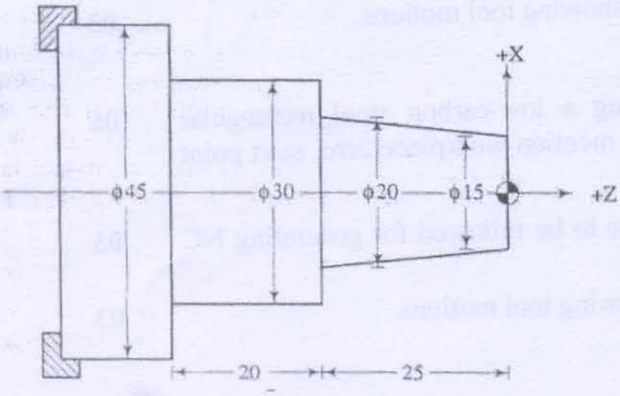

Figure - 1

G00 Rapid traverse
G01 Linear interpolation
G02 Circular interpolation CW
G03 Circular interpolation CCW
G78, G79 Mill cycle
G80 Canned cycle cancel
G84 Threading cycle
G90 Absolute mode
G91 Incremental mode
G92 Presetting

M00 Program stop
M02 End of program
M03 Spindle CW
M04 Spindle CCW
M05 Spindle OFF
M06 Tool change
M08 Coolant ON
M09 Coolant OFF
M30 End of tape